

Oliver Leask

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EDUCATION

St Catherine's College, University of Oxford

2020 –

- BA Physics: First class honours (80% average)
- MMathPhys Master of Mathematical and Theoretical Physics: First class honours (80% average), in particular: 94% mark in Quantum Information.

2024

MMathPhys subjects included: condensed matter & topological phases of matter, groups & representation theory, QFT, and a dissertation on lattice gauge theories.

Nottingham University Academy of Science and Technology

2018 –

A-levels: 4 A*'s in Maths, Further Maths, Physics, and Chemistry.

2020

RESEARCH EXPERIENCE

MMathPhys Dissertation, Department of Physics, University of Oxford

2024

Supervisors: Prof. Alessio Lerose & Dr. Jasmine Brewer

- Reviewed literature on lattice gauge theories, before creating from scratch a 1-D simulation of the collision dynamics between meson-like quantum states, using Python.
- Compared the computed collision spectra against those predicted by theory, and produced original results on real-time dynamical string-breaking, where one meson splits into several.
- Analysed shortcomings of the string-breaking model, and described how to perturbatively obtain an effective Hamiltonian which would allow more accurate results.

Research Assistant, Department of Chemistry, University of Oxford

2023

Supervisor: Prof. Peter Hore

- Worked on a 6-week solo computational project in atomic physics / spin chemistry. It focused on the processes thought to allow migratory birds to navigate using geomagnetic fields.
- After reviewing existing literature, I started from a toy model in Mathematica, and used Liouville space methods to improve its scope and computational efficiency.
- Participated in meetings, presented my findings to the group and in a short timeframe, improved on existing methods and laid the groundwork for a related master's project the following year.

PROFESSIONAL EXPERIENCE

Full Stack Developer, Calypso Commodities, Berlin

10/2024 –

- Employed full-time as a developer for a software startup company, dealing with mathematical optimisation of maritime commodity trading.
- Mostly independently, I have conceptualised, planned and implemented many weeks- or months-long technical projects, requiring work on all levels of the software architecture, while meeting tight deadlines. This has given me an excellent combination of work ethic, organisational skills, and improved logical thinking.
- Frequently presenting new features to audiences of varying levels of expertise.

Present

SKILLS & ACCOMPLISHMENTS

Neville Robertson Prize

2023

Awarded for best undergraduate exam performance in St Catherine's College, in physics.

Oxford University Physics Society Media Officer

2024

Was involved with the physics community at Oxford, as media officer of the physics society. I helped organize, film & photograph all lectures & inter-university events.

DipLCM in Music Performance

2019

Diploma in piano performance from the London College of Music.

Head Boy of Sixth Form

2019

Enjoyed several opportunities to improve my presenting and public speaking skills.